

in association with the video footage is data indicative of content identified by the object detection module 154.

The object detection module 154 begins by identifying objects of interest in a given image frame 501. Objects of interest may be identified based on feature learning techniques. Methodologies for identifying relevant features of objects may be, but are not limited to, Shi-Tomasi corner detection algorithms or Harris Corner Detection Method, as would be understood by one of ordinary skill in the art. With an object identified in the image frame 501, the object detection module 154 leverages a cascade of neural networks 507, 509 to identify aspects of the object.

For example, if the object class of the identified object is that of a human being, then the object detection module 154 may further analyze the image 501 using a neural network module 507B configured to identify different classes of people (male, female, race, etc.). Subsequently, with aspects of the person identified, the object detection module 154 may leverage other, more detailed neural networks 507, 509 to further identify aspects of the person such as, but not limited to, a neural network 509D for identifying clothing types (jacket, pants, shorts, hat, etc.) and a neural network 509ZZ for gauging height and weight.

As other examples, a neural network 507A may be assigned to determine if an image 501 contains a vehicle class of objects. Another neural network 507C may be assigned to determine if an image 501 contains a bicycle class of objects. Similarly, another neural network 507D may be assigned to determine if an image 501 contains an animal class of objects. As understood by one of ordinary skill in the art, other classes of objects are possible and are included within the scope of this disclosure.

For these class objects, other neural networks 509A, 509B, 509C, 509D, and 509ZZ may be assigned to determine if each class of object found by the class neural networks 507 have aspects, such as, but not limited to, make/model of vehicles; colors; license plate readers/recognition modules; clothing types; and/or height/weight of people objects, just to name a few. As understood by one of ordinary skill in the art, other aspects for each class of objects are possible and are included within the scope of this disclosure.

With the outputs from the relevant neural networks 507/509, the object detection module 154 may work to store the outputs in database 120 in association with the image frame 501 (and, by extension, the video footage segment that contains the image frame 501) along with the time and location data. In this way, the database may be configured to be leveraged by the historical analysis module 162 and/or search module 156 and/or alert triggering module 158 as previously described.

Referring now to FIG. 5B, this figure is a logical flow chart for a method 507', 509' for training a neural network 507, 509 to detect objects in images 501. This method is usually performed by the DSOQ server 103. Once the method 507', 509' is complete, the server 103 may send the calculated weights for the one or more neural networks [acting as filter(s)] to the object detection module 154 of FIG. 5A.

One goal of the machine-based object detection method 507', 509' as illustrated in FIG. 5B is to be able to input an image (Step 510) and output a label of the objects in that image (e.g. car, dog, etc.) (Step 540). Here, the method 507', 509' may train each neural network 507, 509 of FIG. 5A its particular class or aspect as mentioned above in connection with FIG. 5A. A common technique for creating an object detection system is to train one or more convolutional neural

networks via machine learning, where the model's accuracy is improved by being provided examples of correct results.

To accomplish this, a dataset must first be compiled with example images where the objects in the image are labeled prior to training (Step 510). All objects that are desired to be detected must be known and incorporated into the model and example data beforehand. Many existing datasets are available for use, such as COCO and Open Images; otherwise, one may label his/her own images. For training, it is recommended to have at least 50,000 images/video segments of each object to be detected.

With this dataset in hand, a convolutional neural network may be trained to learn how to correctly detect objects in an image. As understood by one of ordinary skill in the art, a neural network usually includes a graph of matrix operations where some matrices have adjustable variables (i.e. weights). One goal of the training process is to be able to adjust these weights so that certain input values (i.e. images) result in certain output values (i.e. correct labels). Existing network architectures exist, such as Faster RCNN and YOLO, and are at least two preferred architectures as of this writing.

The training process for a neural network may include inputting labeled images/video segments into a neural network (Step 510), evaluating the image through the neural network (Step 515), and comparing the predicted label generated by the neural network with the actual label (=label made prior to training) (Decision Step 520).

If there is a difference between the expected and actual values (i.e. loss) as calculated in decision Step 520—where the prediction is wrong/off—the method goes to step 525. If there is no difference between the expected and actual value (i.e. no loss), then the method proceeds to decision Step 530.

In Step 525, the difference between the expected and actual values is calculated and applied to the matrix weights of the neural network such that the loss is reduced for the next iteration of Step 520, improving the accuracy of the neural network.

After Step 525, in decision step 530, it is determined if a desired total accuracy of the neural network has been achieved. If the inquiry to decision Step 530 is positive—meaning that the desired accuracy was achieved in Step 520, then the “Yes” branch is followed to Step 535.

If the inquiry to decision Step 530 is negative—meaning that the desired total accuracy based on the comparison in Step 520 has not been achieved, then the “No” branch is followed back to Step 510 wherein the next training image in the training dataset is inputted into the neural network. In Step 535, the network matrix weights are saved. In step 540, the saved network matrix weights may be used to predict labels for objects 146 (i.e. license plates, cars, people, etc.) in new images (not from the training dataset).

This process is then repeated (returned to Step 510) across the entire dataset until the weights begin to converge. A neural network 507, 509 with converged weights provides consistent accuracy across the entire dataset, yielding an object detection model in which you can input a new image and detect the objects in that image with an expected accuracy.

Once the neural network 507, 509 achieves a desired, selected accuracy, the DSOQ server 103 may transmit the saved network matrix weights to the object detection module 154 so that the one or more neural networks 507, 509 that form the object detection module 154 may be updated.

For a specific neural network training example, see a journal article entitled, “Faster RCNN: Towards Real-Time Object Detection with Region Proposal Networks,” authored